

✓ **Correct** 1/1 Points

22

Why do Mail servers use SMTP handshaking in application layer although there is a TCP handshaking in the transport layer? 

to check if the email is sender is not blocked and ok to send
and to check if the receiver mail is in the mail server

✓ **Correct** 1/1 Points

20

List the components of cookie technology.

- 1- cookie header in response message
- 2- cookie header in request message
- 3-cookie file in the user"s end system
- 4- back-end database

✗ **Incorrect** 0.5/1 Points

19

Which services can be granted and can't be granted by TCP when it uses flow control and congestion control? (Justify your answer) 

TCP only grants reliability but does not grant security or time or throughput

✓ **Correct** 2/2 Points

17

Describe two advantages and two challenges of peer-to-peer application architecture.

the advantages : Scalability ,low cost
the challenges : the Security ,make the peers
contribute with other

✓ **Correct** 2/2 Points

16

Consider an institution has an access line of 15 Mbps. The access line services 20 requests per second. Suppose that the average size of a request is 1Mbits. Consider that the Internet delay is 2 sec; LAN delay is 50 msec. What is the average delay experienced by requests when using proxy server with hit rate =20%? (justify your answer).

Traffic intensity on access link = $(20 * 1 * 10^6 * 0.8) / 15 * 10^6 = 1.0667$
traffic intensity > 1 ----> infinity delay (can't determine the average delay)

✓ **Correct** 1/1 Points

15

Why a Web cache is considered both a server and a client at the same time? 

it send the requested messages to the clients so it acts like a server
and it request the pages from the main server to cache it o it acts as a client

✓ **Correct** 1.5/1.5 Points

12

Consider that there are 20 servers, and 20 clients connected to the core of the computer network. Assume that there are 20 simultaneous downloads taking place, involving 20 client-server pairs. Suppose that these 20 downloads are the only traffic in the network at the current time. Consider that there is a common link with transmission rate of 5Mbps in the core traversed by all 20 downloads. Suppose that every server access link has rate of 2 Mbps; every client access link has rate of 1Mbps ; the transmission rates of other links in the core except the one common link are much larger than 5 Mbps. What is the end-to-end throughput for each download? (Answer with justification). 

it will be the smallest rate which is $(5/20) = 0.25\text{Mbps}$

✓ **Correct** 2.5/2.5 Points

11

Consider that there are two hosts (A & B) connected by 2 packet-switches and three links; every link has transmission rate of R bps. Consider that host (A) sends 4 packets to host (B) back-to-back; each packet has length of L bits. Let d_i and s_i , for $i = 1, 2, 3$, denote the length and the propagation speed respectively of link i starting from host(A). Ignore the processing delay and find with justification the time that the 4 packets take to reach Host (B) in case of one of the packet-switches uses store and forward and the other one uses cut-through.

$$L/R + d_1/s_1 + L/R + d_2/s_2 + d_3/s_3$$

$$L/R + L/R + d_1/s_1 + L/R + d_2/s_2 + d_3/s_3$$

$$2L/R + L/R + d_1/s_1 + L/R + d_2/s_2 + d_3/s_3$$

$$3L/R + L/R + d_1/s_1 + L/R + d_2/s_2 + d_3/s_3$$

14

Consider a sender can transmit at a rate of 1Mbps bits/sec in both directions. Suppose packets containing only control (e.g., ACK or handshaking or request) are 100 bits long. Consider the HTTP protocol, and suppose that the initial downloaded object contains 3 referenced objects from the same server. Suppose that each downloaded object is 100 Kbits long except the last one which has 200Kbits. Let RTT denotes the round trip time between the local host and the server containing the objects. Find with justification the total time needed to download the objects assuming parallel persistent HTTP.

$$3RTT + (3 \cdot 100 / 10^6) + (100 \cdot 10^3 / 10^6) + (100 / 10^6 / 3) + (200 \cdot 10^3 / 10^6 / 3)$$